Dated:

Predicted Rain

Beenish Mamnoon 1505 2041

	OL	T	H	W	R
1.	Sunny	Hot	High	Weak.	NO.
2.	Sunny	Mild	High	Strong	NO.
3.	Cloudy	Hot	High	Weak.	Yes
4.	Rainy	Mild	High	Weak	Yes
5.	Rainy	Cool	Normal	weak.	Yes.
6.	Rainy	Cool	Normal	Strong	NO.
7.	Cloudy	Mild	Normal	Strong	Yes
8.	Sunny	Cool	Normal	Weak	NO.

Entropy Total no of rocus = 8.
" " " Yes = 4
" " " NO = 4

Temp

$= -\frac{4}{8}\log(2)\frac{4}{8} - \frac{4}{8}\log(2)\frac{4}{8} =$ } Hot

$-\left(\frac{4}{8}(0.15) + \frac{4}{8}(0.15)\right) = 0.$

Hot

Sunny $= -\frac{0}{3}\log(2)\frac{0}{3} - \frac{3}{3}\log(2)\frac{3}{3} = 0$ } ~~mild~~.

Mild

Cloudy $= \frac{1}{2}\log(2)\frac{1}{2} - \frac{1}{2}\log(2)\frac{1}{2} =$

$-0.075 - 0.075 = -0.1505$ } ~~cool~~.

~~Entropy~~ $= 0 - \frac{0}{8} =$

Rainy = Cool

Dated: Temperature

Hot $= -\frac{1}{2}\log(2)\frac{1}{2} - \frac{1}{2}\log(2)\frac{1}{2}$

$= -0.1505$.

Mild $= -\frac{2}{3}\log(2)\frac{2}{3} - \frac{1}{3}\log(2)\frac{1}{3}$.

$= 0.133 - 0.033 = -0.463$

Cool $= \frac{1}{3}\log(2)\frac{1}{3} - \frac{2}{3}\log(2)\frac{2}{3} =$

$0.033 - 0.133 = -0.1$.

$$0 - \frac{8}{8}(-0.1505) - \frac{3}{8}(-0.463) - \frac{3}{8}(-0.1)$$

Temp = 0.248.

Hot = −0.1505 ✓
mild −0.463
cool −0.1

(2) IG of Humidity.

High ~~Hot~~ $= -\frac{2}{4}\log(2)\frac{2}{4} - \frac{2}{4}\log(2)\frac{2}{4}$

$= -0.075 - 0.075 = -0.15$.

Normal $= -\frac{2}{4}\log(2)\frac{2}{4} - \frac{2}{4}\log(2)\frac{2}{4} = -0.15$

$0 - \frac{4}{8}(-0.15) - \frac{4}{8}(-0.18)$ $= 0$

~~0 + 0.15~~

Dated:

Date

~~Heat~~. Windy.

$$-\frac{3}{5}\log(2)\frac{3}{5} - \frac{2}{5}\log(2)\frac{2}{5}$$

$$-0.18-$$